MIL-F-52531B(ME)
25 May 1983
SUPERSEDING
MIL-F-52531A(ME)
21 January 1975

MILITARY SPECIFICATION

FLOTATION UNIT, PIPELINE

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an air-filled vessel used to float a concrete-coated pipeline into position prior to submersion.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

STANDARD

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-889	- Dissimilar Metals.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3835

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2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein.

DRAWING

ME

TA13213E7907 - Flotation Unit, Pipeline Assembly

(Copies of standards and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. The issue of the document which is indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARD INSTITUTE, INC (ANSI)

ANSI/ASME - Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Applications for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The flotation unit shall be in accordance with Top Assembly TA13213E7907 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly and operation of the flotation unit. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained

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from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Material. Material shall be as shown on the drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.6).

3.3.1 Material deterioration and control. The unit shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environment to which the item may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.4 Workmanship.

3.4.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. Burned surfaces of flame-cut material shall be ground or machined sufficiently to remove ash and cooling checks. The interior surfaces of the flotation unit shall be cleaned of all loose scale and other foreign material before the tank heads are welded on.

3.4.2 Welders and welding. Before assigning any welders to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by the following listed code for the type of welding operations to be performed and that such qualification is effective as defined by the code:

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ANSI/ASME, Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

3.4.3 Welding. The surfaces to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Parent materials, weld filler metals and fabrication techniques shall be as required to enable the flotation unit to conform to the examination requirements specified in section 4. Welds shall develop adequate strength to transmit stress to the parts connected when the flotation unit is tested to hydrostatic pressure of 125 pounds per square inch (psi). All welds shall be airtight and watertight. Peening shut pin holes to stop leakage shall not be allowed. Welding shall be done by the shieled electric-arc method.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and drawings, as applicable.

4.1.2 Disassembly inspection. Failure of any examination or test by the first article sample, shall be cause for disassembly, in the presence of a Government representative, of the first article sample to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first article sample. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

4.1.3 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the quality assurance provisions (QAP) shown on the drawings. The drawings specify the characteristics requiring QAP

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inspection, the sampling plan and the basis for acceptance and rejection (see 6.4 and 6.5.1).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article flotation unit shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Test. The first article flotation unit shall be tested as specified in 4.5.2. Failure of this test shall be cause for performing the inspection in 4.1.2.

4.4 Quality conformance inspection.

4.4.1 Examination. Each flotation unit shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Test. Each flotation unit shall be subjected to the test specified in 4.5.2. Failure of the test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The flotation unit shall be examined as specified herein for the following defects:

Major

101. Any part or component not in accordance with QAP requirements as shown on the drawing.
102. Material not as specified.
103. Materials are not corrosion resistant or treated to be made corrosion resistant for the applicable storage and operating environment.

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104. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
105. Contractor does not have documentation available for identification of material, material finishes or treatments.
106. Workmanship not as specified.

4.5.2 Weld test. Each flotation unit shall be tested to an internal hydrostatic pressure of 125 psi. Evidence of leakage shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Marking. Each unit shall be examined to insure that marking is legible, correct and complete.

5. PACKAGING

5.1 Preservation and packing (see 6.7). The flotation units shall be shipped with no additional preservation and packing required.

5.2 Marking (see 6.7). Flotation units shall be marked for shipment and storage in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The flotation unit is intended for use in launching concrete-coated submerged pipelines. Flotation units are strapped to the pipeline on the ocean floor until it is pulled into its final position, at which time the flotation units are released.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When a first article is required for inspection and approval, and numbers of flotation units required (see 3.2).

6.3 First article. When a first article inspection is required, the item will be tested and should be a preproduction model. The first article should consist of one or more flotation units. The contracting officer should include specific instruction in acquisition documents regarding arrangements for examinations, tests, and approval of the first article.

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6.4 Quality assurance provisions (QAP). The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

6.5 Definitions.

6.5.1 Quality assurance provisions (QAP). QAP is a contractural requirement that supplements section 4 of this specification. A QAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.6 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

6.7 Packaging. For the purpose of preservation/packing level marking, the flotation units shall be marked A/A.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-52531B(ME)	Flotation Unit, Pipeline

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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